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Brief Analysis of Sounds Using a Smartphone

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In almost any given day, one will stumble upon sounds that are out of the ordinary, distinct enough to capture one's attention. Ordinary sounds in a home are represented by the white-noise-like background created by speaking, moving around, household appliances running, and so forth. The out of the ordinary is the sparkling sound of a knife blade or a fork, the clinking of glasses, or dropping of a coin in a bowl. With the help of a smartphone and an app that can provide basic sound analysis, we can identify *what* makes a given sound to be out of the ordinary and, in the process, learn more about the sounds we hear. It has already been shown, for example, how a smartphone can be used to check resonant frequencies of whistles,[1] or to map a sound field in polar coordinates[2]—to give just a couple of examples. It is our intention in the following to show the wealth of findings when a variety of sounds from around the house are recorded and analyzed using a smartphone and a readily available app.[3]

The app used for this purpose provides time evolution of the frequency and intensity of a sound recorded through the phone's microphone. The dataset of frequency and intensity vs. time is called a spectrogram. The specific app settings used predominantly are summarized in Table I. Without access to the actual recorded and processed data, saved screen images are used instead. Most of the images show frequency in the Hz to few- KHz range, and within timeframes typically 5 to 14 s long.

To calibrate the reader, in Fig. 1 we show a spectrogram of (a) a quiet house at night along with the middle C on a piano struck twice, and (b) a spectrogram of an E tuning fork (329.6 Hz). The tuning fork was used to check the accuracy of the app in terms of frequency identification. The frequency range shown for the house spectrogram extends to about 4 KHz while the one for the tuning fork covers only just over 1 KHz. We note that "quiet" can be qualified as the sound intensity being in the range of -80 dBFS to -70 dBFS.[4] To further calibrate the reader, in Fig. 2 we show a spectrogram for the cabin of an airplane in the case of (a) a loud, badly vibrating airplane, and the case of (b) a quieter, newer airplane. The sound intensity pushes into the -15 dBFS range, which is 60 dB louder than a quiet room. Additionally, distinct vibrations have sound signatures covering the 50 Hz to 3 KHz range as easily observed below, and unfortunately heard at the time.

Table I. Settings used for the sound recording and analysis app.

Sound Recording	
Audio Sample Rate	8 KHz, 16 KHz, 24 KHz
Recording Timeframe	5.1 s, 7.1 s, 14.3 s, 28.5 s
Gain Control	Auto
Sound Analysis	
FFT Order (No. Samples)	11 (2048 samples)
FFT Frames Averaged	4
Filter	Blackman

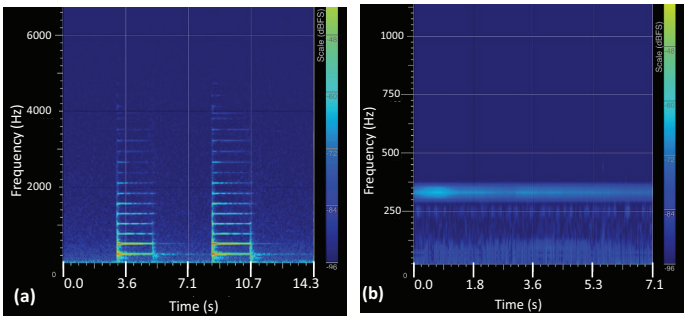


Fig. 1. Spectrograms of a quiet house (a) with middle C on a piano struck twice, and (b) with a tuning fork excited at 329.6 Hz.

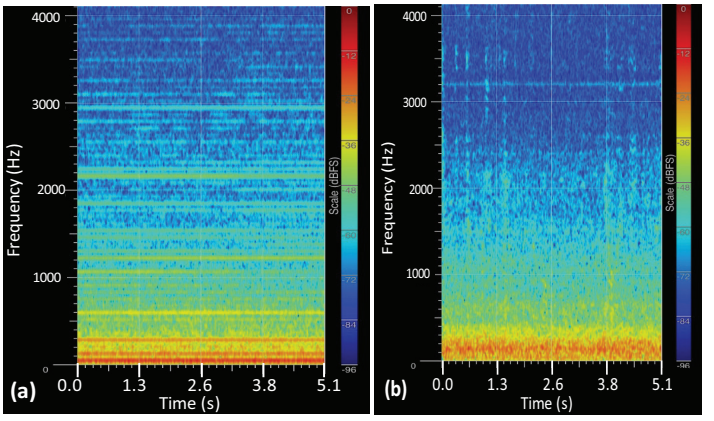


Fig. 2. Spectrograms of an airplane cabin at cruising altitude: (a) noisy airplane, and (b) quieter airplane.

Table II. Characteristic frequencies for a few objects found around the home. We list here up to five of the strongest components in order of increasing frequency.

Object	**Characteristic Frequencies**
Brass tube, doorbell	310 Hz, Fundamental
Glass bowl	820 Hz, 1.9 KHz, 3.5 KHz, 5.2 KHz, 7.3 KHz
Measuring cup	600 Hz, 1.2 KHz, 2.8 KHz, 3.5 KHz, 3.95 KHz
Large metal spoon	400 Hz, 900 Hz, 1.95 KHz, 3.45 KHz, 5 KHz

What makes the piano sound pleasant is not necessarily the cleanliness of its frequency components but the regular spacing of the associated harmonics, and the sound dynamic (the intensity is dropping smoothly providing for dynamic coloring of the sound). This contrasts with the airplane noise, which has components all over the frequency space, while also "adorned" with irregularly spaced resonances of continuously high intensity. The same qualities as those of the piano sound are, not surprisingly, found when studying other typically pleasant sounds as shown in Fig. 3. A brass tube from a doorbell (tube closed at one end), a glass bowl, a metal measuring cup, and a large metal spoon all have beautiful sounds. While the mea-

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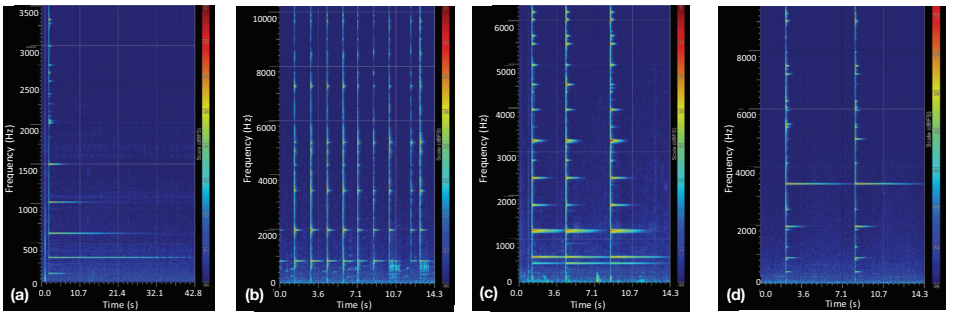


Fig. 3. Spectrograms of (a) brass bell tube, (b) glass bowl, (c) metal measuring cup, and (d) large metal spoon. All objects are excited a few times within the 14-s recording window. The brass tube is excited only once in 43-s data window.

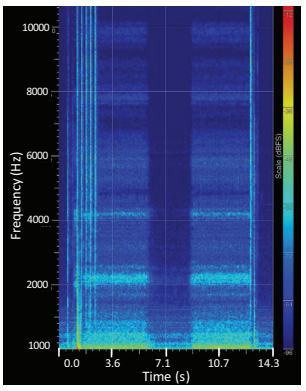


Fig. 4. Spectrogram of an open flame as recorded near the burner of a gas stove. The signatures present in the first 2 s are from the electronic starter.

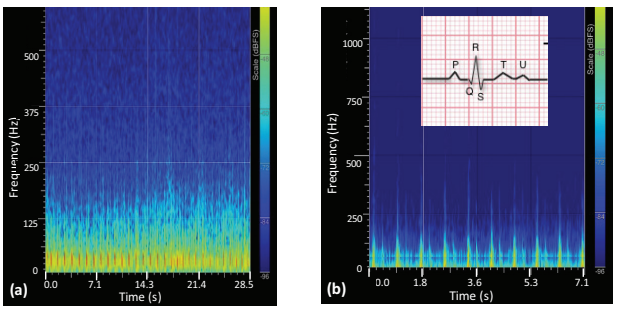


Fig. 5. Spectrogram of a beating heart as recorded just outside the thoracic cavity: (a) Full-screen view (28.5 s long). (b) Zoomed-in view (7.1 s long), and filtered version of (a) with a typical EKG signal superimposed on the image.

suring cup and the spoon were pleasant surprises, what makes the doorbell sound stand out is the time length of its vibrations. The way the objects are held and hit, as well as the recording distance and microphone orientation, will play a role in the observed length of the resonances. Nevertheless, while the bowl or even the piano hold the sound for only several seconds, the brass tube easily reaches into 40-50 s or more.

A variety of other objects/situations have been recorded: forks, glasses, spatulas, metallic shaft of a kitchen mixer, basket of a washing machine, refrigerator, coin falling in a bowl, and so on. We encourage the reader to do the same as it is quite an enlightening exercise. We will highlight below just two other less-usual sounds. In Fig. 4 we are showing the spectrogram of an open flame as recorded near the burner of a gas stove. The succession of identical signatures present in the first two seconds are from the electronic starter, which was fired on purpose several times. The flame is then left on high, turned low, and then back to high, and finally turned off. The frequency "landscape" of the flame shows clearly a somewhat uniform sound emission spectrum up to 10 KHz or thereabouts. There are strong bands in 0 Hz to 1.7 KHz and 2.0 KHz to 2.4 KHz regions, as well as one around 4.2 KHz. More interestingly, there are clearly missing portions in the spectrum, in bands centered around 1.6 KHz, 4.8 KHz, 7 KHz, and 9.2 KHz. Finally, in Fig. 5 we show the spectrogram obtained when the microphone was placed near the heart, just outside the thoracic cavity (the phone is pressed against the skin in the heart area). The beat can easily be recognized and actually measured [Fig. 5(a)]. What is more interesting, with proper filtering the noise can be reduced significantly such that some structure of the actual heartbeat signal can be observed [Fig. 5(b)]. We show superimposed on the spectrogram a typical EKG signal.[5] One has to wonder if the shadow peak observed within each beat period is somehow related to the T-wave from the electrical signal, and if the sound signal acquired can be used as part of a rudimentary medical inquiry.[6]

Based on this type of exploration, a student activity can easily be envisioned for classes dealing with introduction to sound, or classes introducing concepts of sound and sound processing. An example of an activity that can be used as inspiration has already been given elsewhere.[1] The possibility of exploring the world around us with semi-scientific but easily available tools is exhilarating. The simple scientific inquiry described above uses a variety of basic physics concepts while also providing a launching board toward more complex issues such as sound recording and analysis, sounds generated by a plasma, or sound transducers as a medical diagnostic tool. The spectrograms are visually compelling so they can be used aesthetically as well.[6] This should serve as an inspiration for young, inquisitive minds that want to use the smarts from a smartphone.

References

1. Thomas B. Greenslade Jr., "Whistling tea kettles, train whistles, and organ pipes," *Phys. Teach.* **54**, 518 (Dec. 2016).
2. Scott H. Hawley and Robert E. McClain Jr., "Visualizing sound directivity via smartphone sensors," *Phys. Teach.* **56**, 72 (Feb. 2018).
3. For the iOS we have used Oxford SpectrumView 2.2 from Oxford Wave Research. For the Android one can use Advanced Spectrum Analyzer Pro from Vuche Labs, for example.
4. The highest sound level a specific hardware can handle without distortion is 0 dBFS. Additional information on dBFS definition can be found, for example, here: https://www.webel.com.au/taxonomy/term/1013.
5. "Reading EKG," Practical Clinical Skills, https://www.practicalclinicalskills.com/reading-ekg.
6. A strong, regular T-wave signal can suggest increased levels of extracellular potassium, which is a heart condition called hyperkalaemia. See, for example, here: https://lifeinthefastlane.com/ecg-library/basics/t-wave/.
7. This app was used to generate the image of the week published by *Optics and Photonics News* on Dec. 22, 2015. The image can be found here: https://www.osa-opn.org/home/gallery/image_of_the_week/2015/15-12-21_(1)/.

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